

"Analyzing Forensic Science &
Student Success Across the
Crimson Ice"

The ForenGuins

Editorial Board:
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HAVE YOU HEARD THE LATEST GOOD NEWS?



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STUDENT SPOTLIGHT: THE GENETIC PANOPTICON



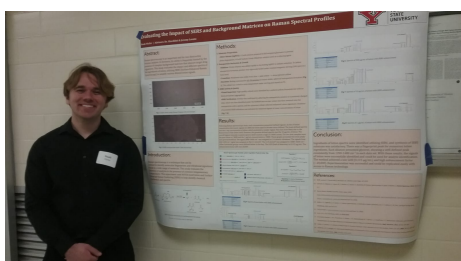
Brandon, Jada, and Megan presenting at
Lahhs-Con | April 9, 2026

On April 9, 2026, three standout forensic students from Youngstown State University, Brandon, Jada, and Megan, took the stage at the Lahhs-Con conference.

Their presentation, "The Genetic Panopticon: Navigating the Ethics of Modern DNA Investigation," offered a deep dive into the evolving ethics of forensic surveillance.

Three cases were discussed by the presenters, including Direct "Trash" Identification (Glenn Raynor v. Maryland, 2014), Familial Search (Lonnie Franklin Jr., "The Grim Sleeper", 2010) and FIGG / Consumer DNA (Joseph James DeAngelo, 2018). The ethical questions to answer included: Does an "Informed Refusal" matter if the state can harvest what you shed involuntarily?, Is "genetic guilt by association" a violation of the 14th Amendment's Equal Protection? and Does one relative's interest in ancestry waive the privacy of an entire family tree?

STUDENT SPOTLIGHT: RAMAN SPECTRAL PROFILES



Noah presenting his poster at YSU Stem
Showcase | April 18, 2026

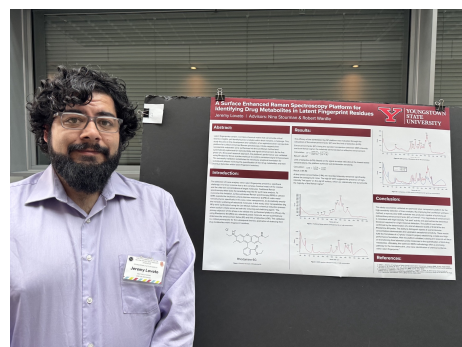
On April 18, 2026, Noah Pfeifer, a talented student from Youngstown State University, presented his research at the YSU Stem Showcase.

His poster, "Evaluating the Impact of SERS and Background Matrices on

Raman Spectral Profiles," explored how surface-enhanced Raman spectroscopy (SERS) and different background matrices influence Raman spectral data.

His research shows that there was a fingerprint peak for numerous lotion ingredients. Each dilution presented glycerol, showing a well-defined peak consistently from 1390-1480 cm^{-3} in each data set. Experimentation was reproduced easily with access to Raman technology.

STUDENT SPOTLIGHT: SAS SYMPOSIUM HONOR



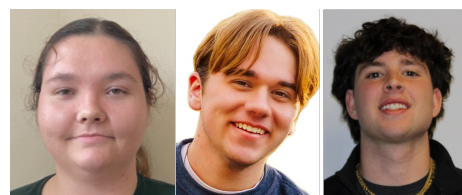
Jeremy Lovato presenting his
award-winning poster research | May 15,
2026

On May 15, 2026, standout student Jeremy Lovato earned prestigious recognition at the New York Capital Region Society for Applied Spectroscopy (SAS) Symposium.

Jeremy won an Honorable Mention for his excellent poster presentation. His award-winning work represents a major achievement in applied analytical spectroscopy, highlighting the dedication and competitive edge of students breaking new ground in laboratory research.

FORMER STUDENT RESEARCH

In Spring semester Savannah Vrtachnik worked on the Italian murder case, Amanda Knox by generating the diorama of the crime scene. Noah Pfeifer worked on the Impact of SERS and Background Matrices on Raman Spectral Profiles and Tyler Canfield worked on privacy issues associated with Forensic technologies.

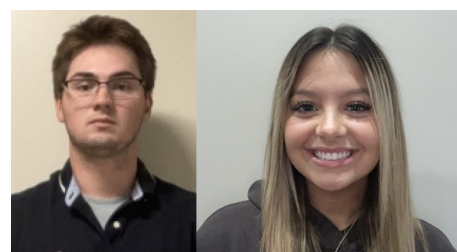


From left to right, Savannah, Noah and
Tyler, three former research students

CURRENT STUDENT RESEARCH

In Fall 2026 Gage Watson will advance fingerprint analysis by identifying specific chemical constituents within latent fingerprint residue. While traditional fingerprinting only maps physical ridge patterns, Gage will utilize Surface-Enhanced Raman Spectroscopy (SERS) to uncover the hidden chemical signatures left behind in sweat and oils. Standard Raman spectroscopy shines laser light on a sample to identify molecules, but the signal from tiny residues is often too weak to detect. SERS solves this by placing the sample on a nanoscale, rough metal surface, which acts as an amplifier to boost the light signal millions of times over.

Arley Lutz will investigate the cognitive mechanisms behind witness testimony to improve the reliability of legal evidence. In the legal system, eyewitness accounts are highly persuasive, yet psychological science shows that human memory does not function like a flawless video recording. Instead, the brain reconstructs past events, making memories highly malleable and susceptible to distortion over time. Arley's research will explore how external factors like high stress, the passage of time, and subtly leading questions from investigators can introduce post-event misinformation into a witness's mind.



Left, Gage researching on SERS and right,
Arley researching on witness testimony

Jeremy Lovato will explore the validity of *Bacillus thuringiensis* as a biological tool for tracking suspects in forensic investigations. This naturally occurring soil bacterium is a close relative of *Bacillus anthracis*, making it a vital, safe model for forensic research. Jeremy's project will focus on using Raman spectroscopy to detect the unique chemical and structural signatures of *Bacillus thuringiensis*, allowing investigators to accurately differentiate it from other closely related bacterial subspecies. Because these microscopic spores cling to clothing and footwear, this advanced detection method can serve as a geolocation tool. By identifying the specific bacterial strains left on a suspect's shoes, forensic teams can scientifically verify the geographic whereabouts of an individual and link them directly to a crime

scene.

UPCOMING CONFERENCES

As the fall semester kicks into gear, several major regional, national, and international forensic science conferences offer premier venues for student research presentations, networking, and professional development. Standout students are encouraged to monitor upcoming abstract submission deadlines.

FEATURED OHIO EVENTS

- **Ohio Identification Officers Association (OIOA)**
(Sept 1–4 | Columbus, OH – Marriott Columbus Northwest)
Focus: The official Ohio state IAI division conference. Excellent regional networking for pattern evidence, crime scene photography, and latent prints.
- **Forensic Lecture Series: Advanced Block**
(Oct 5 onwards | Cleveland, OH – Cuyahoga County Medical Examiner)
Focus: Advanced medicolegal death investigation and case studies.
- **Ohio Academy of Science Annual Meeting**
(Oct 23–24 | Columbus, OH)
Focus: STEM workforce pipeline development, career resource fairs, and research symposia.

NATIONAL & INTERNATIONAL CALENDAR

- **63rd TIAFT Annual Meeting**
(Sept 20–24 | Chicago, IL)
Focus: Forensic toxicology and psychoactive analysis.
- **NEAFS & MAAFS Joint Annual Meeting**
(Oct 4–8 | Pocono Mountains, PA)
Focus: Massive regional criminalistics gathering and student showcases.

LOOKING AHEAD: EARLY 2027

- **79th AAFS Annual Scientific Conference**
(Feb 15–20, 2027 | Orlando, FL – Rosen Shingle Creek Resort)
Theme: "Around the World and Beyond"
Focus: Global forensics, space law, and interdisciplinary fields. Note that general student abstract presentation portals close strictly on August 1.